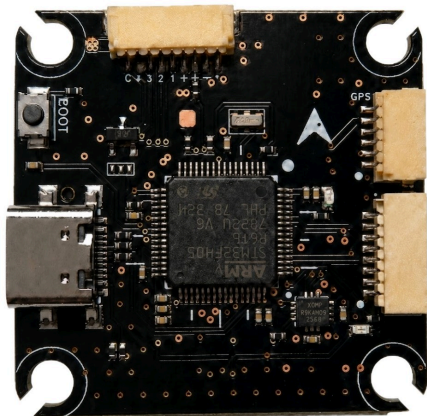
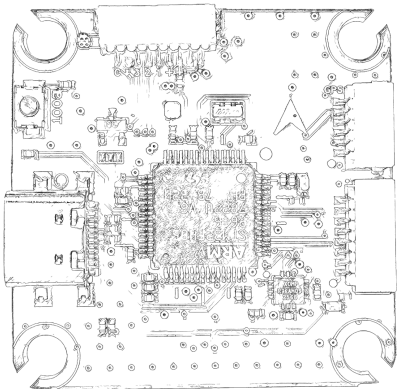


AF-F4 nano v2

User Guide



AF-F4 nano v2 Overview



Compact F4 flight controller with STM32F405, ICM-42688-P and DPS368. GNSS and compass are external, connected through the 6-pin GPS port (USART1 + I2C1).

AF-F4 nano v2 Specifications

• Basic Specifications

MCU	STM32F405
IMU	ICM-42688-P
Barometer	DPS368
GNSS	MAX-M10S — via external GPS module (not onboard)
Compass	QMC5883P — on external GPS module (not onboard)
GPS Port	6-pin connector (UART TX/RX + I2C SCL/SDA)
Operating Voltage	9 – 25 V DC
Output Voltage	3.3V/1A · 5V/3A · 10V/3A
PWM Output	5 channels in current novaX ArduPilot configuration

AF-F4 nano v2 Specifications

• Basic Specifications

Serial Ports	3 hardware UARTs + USB in current novaX ArduPilot configuration
Serial Mapping	SERIAL1 = USART3 · SERIAL2 = USART1 · SERIAL3 = USART2 (USB = SERIAL0)
Blackbox	microSD card
RC Input	PWM / PPM / S.Bus
Size	39.4 × 39.4 mm
Mount Hole	30.5 × 30.5 mm / M4
Weight	9.3 g
Supported F/W	ArduPilot

AF-F4 nano v2 Pinout

• Pinout

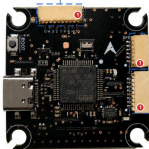
AF-F4 nano v2 Pinout

Each table lists the pins left → right (side connectors: top → bottom) as you see them in the photo.

① = pin 1, the red 1 in its table (FC board) + = power - = GND blue = signal

FC board — top

ESC Interface							
8-Pin - JST-SH - to the 4-in-1 ESC							
8	6	7	5	4	3	2	1
CUR	S4	S3	S2	S1	+	-	-

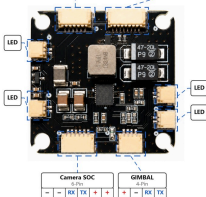


GPS	
6-Pin - JST-SH - USART1 + DC1	
6	-
5	+
4	TX
3	RX
2	SCL
1	SDA

BEC	
8-Pin - JST-SH - to the BEC board	
8	-
7	SCL
6	SDA
5	LED
4	TX
3	RX
2	TX
1	RX

BEC board — connector names as printed by the maker (no pin numbers)

POWER						From FC					
6-Pin - power in						8-Pin - from the FC BEC port					
SCL	SDA	+	+	-	-	-	SCL	SDA	LED	RX	TX



Camera SOC				GIMBAL			
6-Pin				4-Pin			
-	-	RX	TX	+	+	-	-

FC board: MCU STM32F405 - IMU ICM-42688-P - BOOT button and USB Type-C on the left.

AF-F4 nano v2 Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`. Mounting: 30.5 × 30.5 mm / M4.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 9 – 25 V DC.
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-F4 nano v2 Firmware

• Available Firmware

Type	Version	Date
ArduPilot (.apj package)	1.0.12	2026-09-23
Bootloader + App (merged HEX · DFU / SWD)	1.0.12	2026-09-23

Download files from the product page (QR).

AF-F4 nano v2 Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Without the external GPS / compass, position- and heading-based flight modes are unavailable.



<Detailed Manual>

